Program overview

19-May-2025 12:08

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**EWI Electives Service-Education**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
EWI Serviceonderwijs 2024							
IFEEMCS010400	Linear Algebra	5					
IFEEMCS010500	Probability and Statistics	3					
IFEEMCS011100	Calculus for Science, part 1	3					
IFEEMCS011200	Calculus for Science, part 2	3					
IFEEMCS012100	Calculus for Engineering, part 1	3					
IFEEMCS012200	Calculus for Engineering, part 2	3					
IFEEMCS012300	Calculus for Engineering, part 3	3					
IFEEMCS4250	Statistical Learning for Engineers	4					
WI1807TH1-21	Linear Algebra	3					
WI1909TH	Differential Equations	3					
WI2032TH	Numerical Analysis	3					
WI4014TU	Numerical Analysis	6					
WI4019IFO	Nonlinear Dynamical Systems	6					
WI4062TU	Transport, Routing and Scheduling	3					
WI4525TU	Monte Carlo Simulation and Stochastic Processes	5					

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**EWI Electives Service-Education**

EWI Serviceonderwijs 2024

IFEEMCS010400	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	6/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Summary	Course email: ifeemcs010400@tudelft.nl (please don't use personal email addresses)	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Colstruction	
Literature and Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Final score: F IF the midterm score M is higher than the score E for the exam AND E >= 5.0 THEN F = 0.25 M + 0.75 E ELSE F = E All grades are rounded to 1 decimal place. Note that the midterm is not mandatory (and cannot be resited). For the resit the midterm is not relevant anymore.	
Permitted Materials during Tests	None	
Test and result scale	1-10	
Co-Instructor	Dr.ir. R.F. Swarttouw	

IFEEMCS010500	Probability and Statistics	3
Responsible Instructor	Dr. R.J. Fokkink	
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	x/x/4/x or x/x/x/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch English	
Expected prior knowledge	Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, you will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use a computer program to perform elementary simulation and data analysis	
Education Method	Interactive lectures, pre-lecture videos, exercises	
Literature and Study Materials	Book: Mathematical introduction to Probability and Statistics, Dekking et al.	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher.	
Permitted Materials during Tests	1) Formula sheet to be used for exams of Probability and Statistics 2) Any kind of calculator	
Remarks	Please note that this course is available in Q3 and in Q4.	
Co-Instructor	Dr. R.J. Fokkink	
Co-Instructor	Dr. C. Kraaikamp	

IFEEMCS011100	Calculus for Science, part 1	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Course Contents	The goal of the course is to lay a strong foundation in calculus of functions in one variable. We will discuss various techniques to differentiate, integrate, and solve differential equations. Applications in different fields will be discussed. Both a deep knowledge of the theoretical foundations and accuracy in computations will be important.	
Study Goals	At the end of this course, the student will be able to - Use differentiation for linearising a function and apply differentials. - Perform vector arithmetic (addition, scalar multiplication, dot product, cross product). - Work with functions and related concepts (domain, range, graphs of functions, injectivity,etc.). - Perform the basic operations of calculus (differentiation, product laws, chain rule, etc.). - Apply integration techniques for functions of one variable (substitution rule, integration by parts). - Solve simple first-order differential equations by analytical methods. - Draw and interpret slope fields - Do arithmetic with complex numbers, both in Cartesian and polar form (addition, multiplication, division, powers, etc.). - Use complex numbers, perform complex arithmetic in both cartesian and polar form.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

IFEEMCS011200	Calculus for Science, part 2	3
Responsible Instructor	Dr. N.D. Verhulst	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Science, part 1 or WI1708TH1 (EE/AE).	
Course Contents	The course starts with the introduction of a method to solve general second order linear differential equations. But the main focus will be on series. Convergence and absolute convergence of series takes pride of place. Taylor series will be introduced and used to solve differential equations and initial value problems. A start will be made with multivariate calculus. In particular, continuity, limits, partial derivatives, and linearizations of multivariate functions will be discussed.	
Study Goals	At the end of this course, the student is able to - Solve homogeneous and non-homogeneous second order differential equations. - Recall and apply various tests for (absolute) convergence of series. - Give precise definitions and relevant theorems about a.o. alternating series, power series, binomial series, etc. - Compute Taylor series and Taylor polynomials. - Solve differential equations using series. - Find limits and partial derivatives of multivariate functions. - Compute Tangent planes, linearizations, and differentials for multivariate functions.	
Education Method	Colstruction	
Assessment	One written exam at the end.	
Co-Instructor	Dr. C.O. Smet	

IFEEMCS012100	Calculus for Engineering, part 1	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programma students are applying for.	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012200	Calculus for Engineering, part 2	3
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	x/4/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch English	
Co-Instructor	Dr.ir. N.P.K. Chan	

IFEEMCS012300	Calculus for Engineering, part 3	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Expected prior knowledge	Calculus for Engineering 2 (IFEEMCS012200) or a similar course.	
Course Contents	Line integrals and surface integrals. Integral theorems of Green, Stokes and Gauss. Sequences and series. Power series and Taylor series.	
Study Goals	After this course the student should be able to 1. find parametrizations of curves and surfaces (in 2D and 3D); 2. calculate a line integral or a surface integral of a function or a vector field; 3. mention the physical interpretation of these line and surface integrals; 4. define and formulate the physical meaning of concepts like gradient, curl and divergence; 5. calculate line and surface integrals via one of the integral theorems (Green, Stokes, Gauss) 6. determine whether a sequence is convergent or divergent, for example by calculating its limit; 7. determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; 8. determine the interval of convergence of a power series; 9. determine the Taylor series of a simple analytic function, for example sin(x), cos(x), exp(x), ln(1+x), arctan(x), (1+x)^a.	
Education Method	Lectures (per week 2x2 hours).	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	On-campus exams. Partly digital, partly written. See Brightspace for more details. There will be two partial exams, of 90 minutes each: - The first partial exam is about vector calculus; - The second partial exam is about sequences and series. The final grade is the average of the grades for the two partial exams. There is only one resit, about the complete course, that lasts 180 minutes. It is NOT possible to resit only one partial exam.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators allowed.	
Co-Instructor	Dr. F.J. van de Bult	

IFEEMCS4250	Statistical Learning for Engineers	4
Responsible Instructor	Dr. H.N. Kekkonen	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra, Calculus, Probability& Statistics on the level of TU Delft Bachelor; some programming skills in Python.	
Course Contents	Multiple linear regression, logistic regression, regularisation, cross-validation, tree based methods, and short introduction to deep learning.	
Study Goals	After this course you should be familiar with several commonly used statistical and machine learning methods. know how to estimate unknown parameters and make predictions based on data. be able to apply statistical learning techniques using Python. know how to evaluate model accuracy and be aware of the strengths and limitations of different models.	
Education Method	Lectures and exercise + computer labs	
Assessment	85% Final exam and 15% homework assignments. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Enrolment / Application	Please note: students do not need to register via Osiris for this course, only for the corresponding exams. Students only have to enroll via Brightspace.	
Co-Instructor	Dr. N. Parolya	

WI1807TH1-21	Linear Algebra	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch English	
Expected prior knowledge	See the requirements of the specific MSc programme students are applying for.	
Course Contents	- Systems of linear equations, elementary row-operations, (reduced) echelon form - Vector equations, spans - Solution sets of (non)homogeneous systems of linear equations - Linear independence - Linear transformations and standard matrices - Matrix algebra (sum, product), inverse of a matrix - Subspaces in R^n, column space and null space - Basis and Dimension - Determinants - Eigenvalues and eigenvectors - Diagonalization	
Study Goals	After successfully completing the course you will be able to: - solve systems of linear equations. - identify if a system of linear equations is solvable, and if the solution is unique. - show the equivalence between a vector equation, a matrix equation and a system of linear equations. - give a geometric description of solution sets of vector equations. - determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - identify if a transformation is linear and determine the standard matrix of a linear transformation. - perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - determine the determinant of a matrix by applying the properties of a determinant. - apply the properties of determinants (product, transpose, inverse). - determine the eigenvalues and eigenvectors of a matrix. - determine the algebraic and geometric multiplicity of the eigenvalues of a matrix. - determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - analyze the case of complex eigenvalues and eigenvectors.	
Education Method	Lectures, 4 hours per week (2x2)	
Literature and Study Materials	- Book: David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216. - Slides on Brightspace. - Prelecture videos. - Online homework assignments (Grasple). - Old exams on Brightspace.	
Books	David C. Lay, Steven R. Lay, Judi J. McDonald, Linear Algebra and its Applications, Sixth edition, Global Edition, ISBN-13: 978-1292351216.	
Assessment	Digital exam (in Grasple) on campus.	
Remarks	Lectures will be in Dutch! Exams will be in offered in Dutch and also in English.	
Co-Instructor	Dr. E. Emsiz	

WI1909TH	Differential Equations	3
Responsible Instructor	Drs. I.A.M. Goddijn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	Calculus and Linear Algebra	
Course Contents	Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series.	
Study Goals	Learning a number of mathematical techniques necessary for solving various technical-physical model equations.	
Education Method	Colstruction	
Books	Elementairy Differential Equations and Boundary Value Problems, Global Edition ISBN-13: 978-1-119-38287-4 (preferable) or Elementairy Differential Equations and Boundary Value Problems, International Adaptation ISBN-13: 978-1-119-82051-2	
Assessment	Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024.	
Permitted Materials during Tests	Announced during lectures and on Brightspace	
Co-Instructor	Dr.ir. Z.J.G. Gromotka	

WI2032TH	Numerical Analysis	3
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	0/0/0/4 (college) 0/0/0/2 (practicum, gedurende 3 weken)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	See WBMT2049 T2	
Study Goals	See WBMT2049 T2	
Education Method	See WBMT2049 T2	
Assessment	See WBMT2049 T2 and WBMT2049 T3	
Enrolment / Application	Registration for education is not necessary for this course, but the other rules for registration for practicals do apply, see WBMT2049 T3.	
Co-Instructor	D. Toshniwal	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4019IFO	Nonlinear Dynamical Systems	6
Responsible Instructor	Dr. A.B.T. Barbaro	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7.	
Course Contents	An emphasis on modeling using Differential Equations. Dimensional Analysis, Existence and uniqueness of solutions of initial value problems, Autonomous systems, Locally linear systems, Critical points, Periodic solutions, Stability theory, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures. In total 14 lectures of 2 hours each will be given. Following and preparing the lectures takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	To be announced on BrightSpace	
Assessment	The take-home exam consists of 30 exercises. The oral exams will be held at the end of the course, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are done correctly. The take-home exam and the oral exam each count for 50% in the final grading of the course. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. W.T. van Horssen	
Co-Instructor	Dr. H.M. Schuttelaars	
Co-Instructor	Dr.ir. E.G. Rens	

WI4062TU	Transport, Routing and Scheduling	3
Responsible Instructor	Dr.ir. T.M.L. Janssen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we deal with combinatorial optimizations methods for the solution of problems that arise when one has to optimally organize transportation of goods, routing of vehicles, and production schedules. We study, amongst others, the shortest path problem, the assignment problem/transportation problem, the travelling salesman problem, the vehicle routing problem, and the job shop scheduling problem.	
Study Goals	After this course you are able to: - recognize a problem as a discrete linear optimization problem and is able to provide a mathematical formulation for it. - solve the shortest path problem and the transportation problem as well as some small flow shop problems. - solve the travelling salesman problem by the Branch and Bound algorithm. - apply several heuristic solution methods for the travelling salesman problem and the vehicle routing problem. - reproduce basic theorems concerning the mentioned problems and is able to prove some of these theorems. - explain how to solve large scale problems, especially shortest path and vehicle routing problems. - define scheduling problems and complexity theory. - implement the algorithms studied in the course.	
Education Method	Lectures	
Literature and Study Materials	Course notes and handouts (made available via Brightspace).	
Assessment	Written exam and a written resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Elective	Yes	
Tags	Optimalisation Transport & Logistics	
Co-Instructor	Dr.ir. J.T. van Essen	

WI4525TU	Monte Carlo Simulation and Stochastic Processes	5
Responsible Instructor	Dr.ir. L.E. Meester	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	

Expected prior knowledge

Students are expected to have: superior knowledge and skill in elementary probability (details below); experience working with Python, Matlab, or R; and a mathematical inclination. If this does not apply to you and you still want to enroll: contact the instructor.

Good skills in elementary probability are indispensable as they are the foundation for the more advanced material of the course. We build on the concepts, tools, and methods of elementary probability, and students taking this course must be (or rapidly become) black-belt masters in these foundations, specifically: Chapters 2-5, 7, 9, and 10 of Dekking et al; prior knowledge of Chapters 13 and 14 is helpful, though we will revisit these subjects during the course.

Testing yourself and doing a thorough review in the first few weeks is required; we will also spend some time on this in class to help you do this efficiently; after this a diagnostic test (carrying part of the course grade) will assess your mastership; sub par performance means you cannot follow the course.

Course Contents

This course on Monte Carlo simulation and stochastic processes consists of a mixture of these components:
modelling and models (and some of their pluses and minuses),
learning a number of stochastic processes and some of their properties,
doing Monte Carlo simulations,
analyzing the generated data and drawing conclusions from them.

Contents and format of this course are intended to combine the following:
1. real (possibly stylized) problems as starting points;
2. enough about modelling to make participants aware of the impact of modelling decisions;
3. identification of a number of stochastic processes present in/related to models and problems;
4. (limited) treatment of underlying theory, guided by:
   (a) what is necessary to ensure that participants know how to do things right and
   (b) what enables them to do things smartly;
5. facts, properties, and theorems:
   (a) sincere attempt to understand where possible;
   (b) focus on practical implications and uses;
   (c) illustrations rather than proofs or theory;
6. useful tools and techniques from statistics, probability, and stochastic processes;
7. simulation as aid to: obtaining answers, intuitive understanding, illustration of theory;
8. providing as much opportunity as feasible for participants to work on/with problems from their own field.

Tentative topics/contents list:
Review basic probability: most to be done as self-study, some in class, plus a mandatory test (see Assessment). Models, modelling. Simulation: Convert a model to a program and run it. How to expose and locate errors. Data, empirical distribution; estimation. Accuracy, efficiency. Sensitivity. Rare event simulation. Experimental design. Conditional distributions and expectations; conditioning as a tool. Multivariate distributions: general (some examples); multivariate normal. Stochastic processes: Random walk. Poisson process; renewal processes. Markov chains: discrete time; continuous time; long-term behaviour; embedded Markov chain; birth and death process; Brownian motion. A snippet of diffusions, (solutions to) stochastic differential equations (SDEs).

Study Goals

Learning objectives: after following the course, participants will be able to
1. in an iid setting, carry out Monte Carlo simulations; and
2. extract and collect, summarize, and interpret the data thus generated; and
3. assess accuracy/uncertainty in estimates, and solve questions like How many runs for 1% accuracy?;
4. the above (three) also in some non-iid settings, e.g., for a stationary process, using the batch-means method;
5. for small-sized practical problems: build a suitable model and simulate it;
6. extract and collect required model parameters from a problem description or background;
7. recognize where modelling choices are made;
8. make elementary modelling decisions themselves;
9. recognize and describe in mathematical terms (the properties of) a number of stochastic processes, among others: Markov chains, random walks, Brownian motion;
10. recognize/identify the covered stochastic processes in practical settings/problems;
11. use the tools and concepts from probability theory to describe, discuss, and manipulate (the components of) stochastic models;
12. where applicable, solve problems by means other than Monte Carlo, i.e., analytically or numerically, procuring exact answers, by identifying a stochastic process and using its known properties; for example, concerning long-term system-behaviour.

Education Method

Weekly class meeting of two hours. About two thirds spent on lecture; about one third on questions and exercises. A number of so-called "Hands-on" assignments: a collection of exercises that combine the analytical content with computer simulation; to be done individually or with a partner. The participants are expected to perform a significant amount of simulation in Python, Matlab, or R (or other suitable language). Spread over the semester, five homework assignments.

I strive to make a classroom available during the (probably) Friday lunch break, in which the participants can work on their own and where the TA (if there is one) is available for questions. In addition, I will hold a weekly drop-in office hour, most likely at lunch time (details on Brightspace).

Literature and Study Materials

Unfortunately, for this course there is no really appropriate textbook, one that combines modelling, Monte Carlo simulation, and stochastic processes.

For the review of basic probability and statistics, you are referred to Dekking et al, "A modern introduction to probability and statistics," Springer, 2005; the text is very well suited for individual review and practice, as it contains many worked examples, (quick) exercises, answers and full solutions. To be successful in the course, participants must have developed superior skill wrt the tools and techniques in the first half of the book (see "Expected prior knowledge").

Most of the material on stochastic processes can be found in "Probability Models" by Sheldon M. Ross; the book contains many worked examples and exercises; its heuristic and intuitive approach fits that of the course; references will be to the 10th Edition. A slightly more mathematical, but still very accessible reference covering the same area is "An introduction to stochastic modeling" by Howard M. Taylor and Samuel Karlin.

Lecture slides, notes, and additional references will be posted on Brightspace.

	A laptop is required and it is recommended that you bring it to class. Many languages and/or software packages are suitable: I prefer you use the one you are most familiar with. I am migrating to Python, but in the material presented in class and distributed via Brightspace, there still is some Matlab code. You may use a language/tool other than those three, but talk to me first (because you should be able to use your software on the exam and it should be possible to arrange this). In all cases: you should have the software installed and be familiar with its use.
Assessment	1. Probability test on knowledge and skills needed in/for the course (15%); passing score must be obtained to take the course. 2. Assignments (five; each counts for 4%), which may be done with a (=one) partner. 3. Final exam (on-campus): closed-book computer exam (with cheatsheet allowed, specs on Brightspace); counts for 65%. To pass the course, the probability test must be passed and the final exam grade must be 6 or higher; course grade is weighted average.
Permitted Materials during Tests	To the final exam you may bring a self-compiled formula sheet: one A4 page with normal-sized text on about 25 lines (or any reasonable equivalent of this), made by you, of things you would find helpful.
Elective	Yes
Tags	Adventurous Diverse Mathematics Modelling Practicals Stochastics
Co-Instructor	Dr.ir. G.N.J.C. Bierkens

Dr. A.B.T. Barbaro

Department	Mathematical Physics
Telephone	+31 15 27 81211

Dr.ir. G.N.J.C. Bierkens

Department	Statistics
Telephone	+31 15 27 84593
Room	36.HB 06.300

Dr. N.V. Budko

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 86050
Room	36.HB 03.050

Dr. F.J. van de Bult

Unit	Elektrotechn., Wisk. & Inform.
Department	Analysis
Telephone	+31 15 27 82541
Room	36.HB 08.300

Dr.ir. N.P.K. Chan

Department	Statistics
Room	36.HB 06.130

Dr. E. Emsiz

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 84579
Room	36.HB 07.060

Dr.ir. J.T. van Essen

Unit	Elektrotechn., Wisk. & Inform.
Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 86266
Room	36.HB 04.280

Dr. R.J. Fokkink

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 89215
Room	28.1.E300

Prof.dr.ir. M.B. van Gijzen

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 82519
Room	36.HB 03.060

Drs. I.A.M. Goddijn

Department	Numerical Analysis
Telephone	+31 15 27 86408

Department	Numerical Analysis
Telephone	+31 15 27 86408

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 86408
Room	36.HB 03.260

Dr.ir. Z.J.G. Gromotka

Department	Mathematical Physics
Telephone	+31 15 27 81887

Dr.ir. W.T. van Horssen

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 83524
Room	36.HB 05.080

Dr.ir. T.M.L. Janssen

Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 83882
Room	36.HB 04.070

Dr. H.N. Kekkonen

Department	Statistics
Telephone	+31 15 27 88337
Room	36.HB 06.060

Dr. C. Kraaikamp

Department	Applied Probability
Telephone	+31 15 27 81910

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 81910
Room	36.HB 07.300

Dr.ir. L.E. Meester

Department	Applied Probability
Telephone	+31 15 27 87258

Unit	Elektrotechn., Wisk. & Inform.
Department	Applied Probability
Telephone	+31 15 27 87258
Room	36.HB 07.320

Dr. N. Parolya

Department	Statistics
Telephone	+31 15 27 88346
Room	36.HB 06.290

Dr.ir. E.G. Rens

Department	Mathematical Physics
Telephone	+31 15 27 83795
Room	36.HB 05.310

Dr. H.M. Schuttelaars

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 83825
Room	36.HB 05.270

Dr. C.O. Smet

Department	Statistics
Telephone	+31 15 27 88666
Room	36.HB 06.260

Dr.ir. R.F. Swarttouw

Department	Analysis
Telephone	+31 15 27 83634

Room	36.HB 08.290

D. Toshniwal

Department	Numerical Analysis
Telephone	+31 15 27 88315
Room	36.HB 03.290

Dr. N.D. Verhulst

Department	Discrete Wiskunde en Optimalisering
Telephone	+31 15 27 88620
Room	36.HB 04.070